SDG 3 & SDG 6: A Business Intelligence Analysis for Unilever using Global Water and Health Indicators (2006-2020)

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1.Business Problem Identification & SDG Alignment

Access to clean water, sanitation, and basic hygiene is a major public health is a challenge for economies globally. It is a key factor that impacts the wellbeing, employment, and economic resilience among the entire society. Research backed by the United Nations states that there are around **2.2 billion people that lack safely managed drinking water** and around **3.6 billion people does not have safely managed sanitation systems.** These issues not only threaten the public health but also hinder the economic development of a country, resulting in people suffering from diseases that could have been prevented. These issues are linked directly in impacting the child mortality rate that is increasing due to the lack of sanitation, drinking contaminated water, and waterborne diseases.

Unilever is a British multinational consumer packaged goods company that operates in a landscapes that stretches across diverse socio-economic regions, including the countries that faces severe water scarcity and sanitation challenges. Unilever manufactures products for personal hygiene, home care, beverages and packaged foods, for which water is key ingredient**. Without access to clean and reliable water manufacturing these products are not possible**. Communities facing water deficiency and high disease burdens tend to work less efficiently, which leads to slower growth and reduced purchasing power in the future. Thus, Unilever is interested in SDG 6 (Clean Water and Sanitation) and SDG 3 (Good Health and Well-being) to strategically align with sustainable operations and maintain future market opportunities.

Unilever's agenda strongly aligns with goals of both SDGs by promoting public health campaigns, education on hygiene, and sanitation contributing to improvement of overall community health outcomes. The company uses approaches like **"3+1 approach"** for cooking and **"Fridge Night"** app used to reduce food wastage. Unilever's operations include initiatives to protect water resources, increase usage of regenerative agriculture, and support community-based water infrastructure. This demonstrates a holistic approach of Unilever to closely align with the goals of SDG 3 and SDG 6. The organization strongly commits to contribute towards the initiative **"Water for All by 2030"** by United Nations.

The business problem examined in this report focuses on understanding **how different countries have variations in water access, sanitation conditions, and safely managed water services influence child mortality,** and how BI Tools support Unilever in its mission to accelerate progress towards SDG 3 and SDG 6. This analysis contains data from 2006-2020 across multiple countries (developing vs developed) to explore disparities in basic water access, gaps in safely managed drinking water, open defecation rate and trajectory that shows under five mortality rate due to lack of sanitation, and safely managed drinking water access.

By converting the global SDG indicators into interactive tableau dashboards, this project uses Business Intelligence as a powerful asset-one that helps Unilever pinpoint the regions that should be prioritized for investments, partnerships, and generate social value. The visual analytics created with the help of tableau provide a clear understanding of global water and health challenges. These analytics will help Unilever move from relying on generic assumptions to actionable insights.

This Dashboard contains specific patterns such as regions with water scarcity, countries facing rising disease burdens, and communities where poor infrastructure threatens both health outcomes and economic productivity. These insights allow Unilever to align its product strategies, community programs, and supply-chain with the most urgent needs on the ground and contribute to long-term development while creating competitive and social value globally.

2.Data Quality, Data Preparation & Ethical Considerations

The dataset used for this analysis contains annual country indicators for nine countries across a 15-year period (2006-2020). Indicators includes access to basic drinking water, safely managed water services, open defecation rate, and under-5 mortality rate. The initial dataset required cleaning to ensure the accuracy in Tableau. The data cleaning was done in python using jupyter notebook where missing values were identified and removed, because the null values can be interpreted as values and that can lead to incorrect visualization. Numeric columns were converted to float, time values were converted into integer from string, and categorical columns were standardized. Whitespace inconsistencies and infinite values in calculations were also handled to avoid misclassification of countries within Tableau.

Ethical considerations taken ensures that the analysis avoids misinterpretation**. The report does not claim causation- only correlation and association**. Country-level comparisons play an important role in analysis as it will avoid any kind of unfair judgements or biases. There are many countries that do not report indicators every year, due to which the dashboards could exaggerate trends and mislead comparisons, to avoid risks, missing values were treated transparently. All datasets used are publicly accessible and contains no personal details, making it fully compliant with responsible data practices.

The data quality is sufficient for country level comparisons, geographical visualization and time trend analysis. Once it is cleaned the dataset integrated seamlessly into Tableau, it helped in creating multiple sheets that visualizes all variables efficiently using KPI cards, line charts, a scatter plot, bar graph, and a global map. The availability of indicators that includes time-series and country selection buttons make the dashboard more interactive to compare data over years.

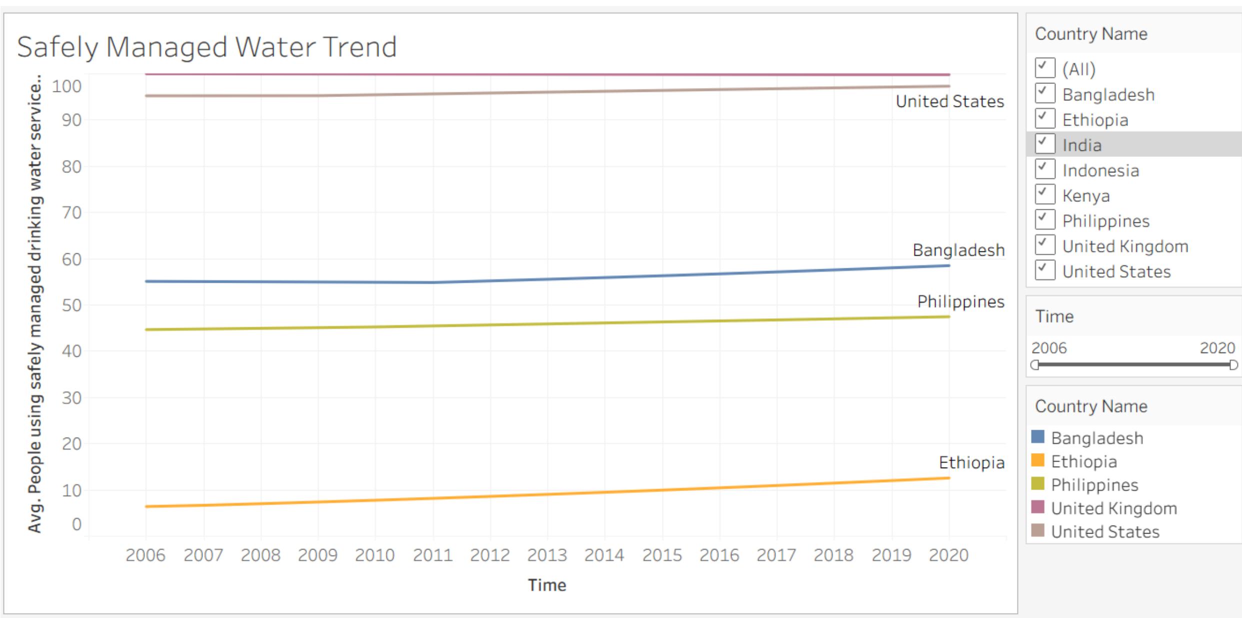
3.Dashboard Design & Insights Interpretation

3.1 KPI Card Insights (2020)

The KPI cards offer an overview of the latest available year (2020) in the dataset, that enables Unilever to quickly gain the perspective on global disparities. Basic drinking water access average **85.73%,** suggesting moderate progress toward **SDG 6.1.** Although the insights about safely managed drinking water access shows **63.14%,** which indicates that nearly one-third of individuals, even with basic access, are consuming water that may not meet safety standards. This gap signifies the systematic weakness in existing water infrastructure.

The average under-five mortality rate of **26.53% per 1000 live births** shows a pattern that children under-five is facing more child health risks mostly in the developing countries. Open defecation, averaging at 6.29%, it is primarily happening in the low-income countries, where sanitation infrastructure is insufficient. This KPI is critical for Unilever, as open defecation contaminates the environment and increases vulnerability to disease which links directly to **SDG 3** which focuses global heath and sanitation.

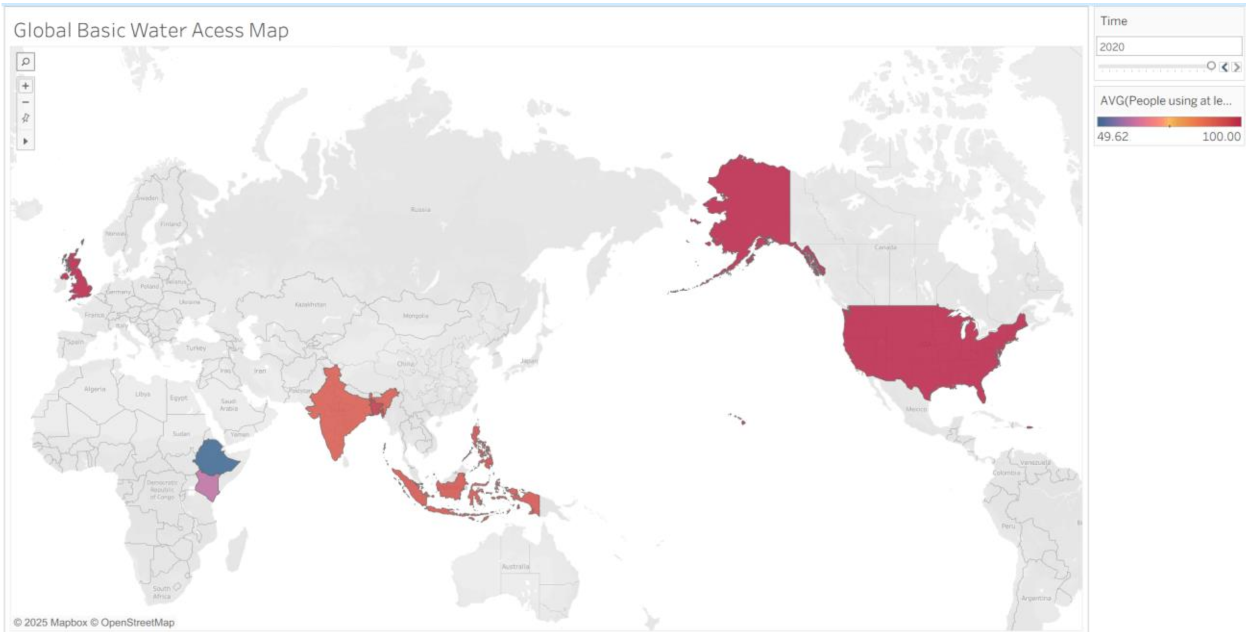
3.2 Safely Managed Water Trend (2006-2020)



The trend for safely managed water over the past 15 years shows a clear division between the developed and developing countries. The United Kingdom and United States of America has access to above 95% safely managed water services, which reflects the established water-treatment infrastructure and high investments in the field of public well-being.

The gap highlights a deeper weakness in national water infrastructure and treatment capacity. The basic water access has increased in all the developing countries in the past 15 years but the facilities to treat water and deliver safely managed water that is safe for consumption is not present. Such deficiencies lead to slower economic growth, limited investment and inadequate resources. To tackle such issues we need more technological advancements and financing structures. Only when all the factors align together can enhance water quality with expanding water access.

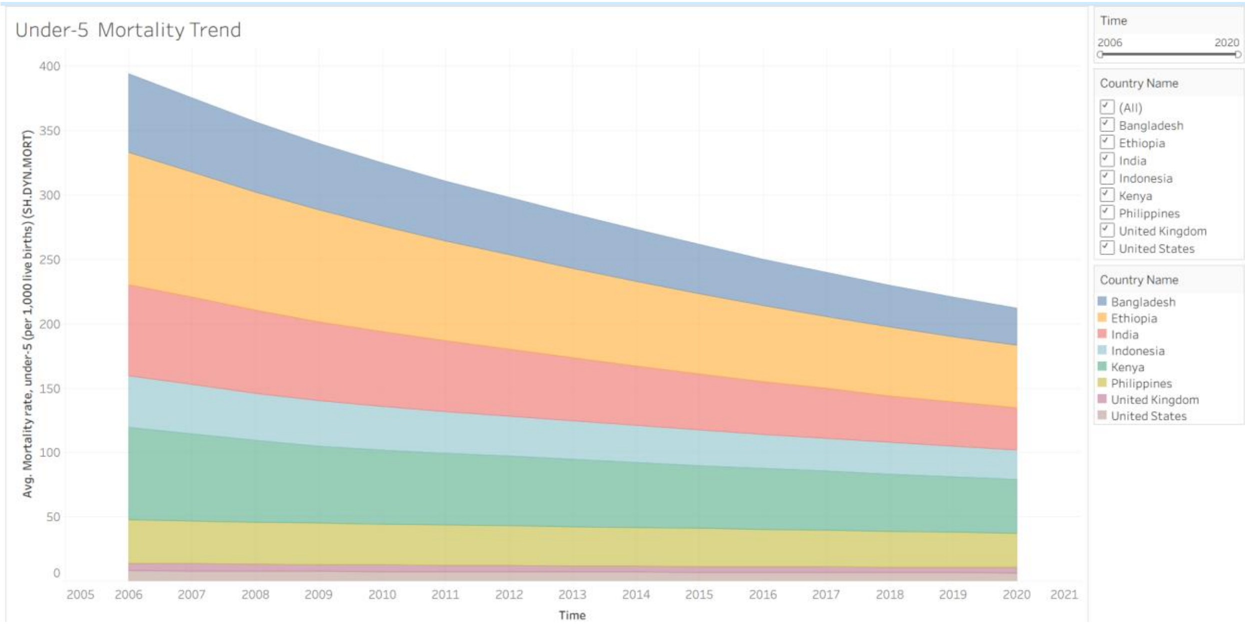
3.3 Global Water Access Map (Basic drinking water)



This Global map represents the clear disparities between the developed and developing nations than any other charts. The regions that are dark-shaded such as **United Kingdom** and **United States of America** shows plentiful supply of clean drinking water **(almost 100%).** However, countries like Ethiopia and Kenya appear very light in comparison of the others. This shows lack of basic drinking water access in African countries.

Through this map Unilever can identify the countries who they should focus more on. Markets with poor water access represent that they need commercial as well as humanitarian needs. Lack of water supply can reduce the demand of water-dependent products such as detergents, home cleaning and personal care items. Thus, strengthening water-security partnerships contributes both to the SDGs and to Unilever's long-term operational stability.

3.4 Under Five Mortality Trend



This chart indicates the declining trend of the under-five mortality rate. It can be seen from the chart that countries like Bangladesh and Ethiopia is improving, although they still maintain higher mortality rates in comparison to UK and USA. India and Kenya also shows steady decline.

This trend is directly proportional to the sanitation services, water quality, and health services. We can also notice that countries with increase in water access declines sharply in child mortality, reinforcing SDG 3.2 target. This decline can increase by educating people about hygiene in high-risk areas and also educate them about water-treatment infrastructure.

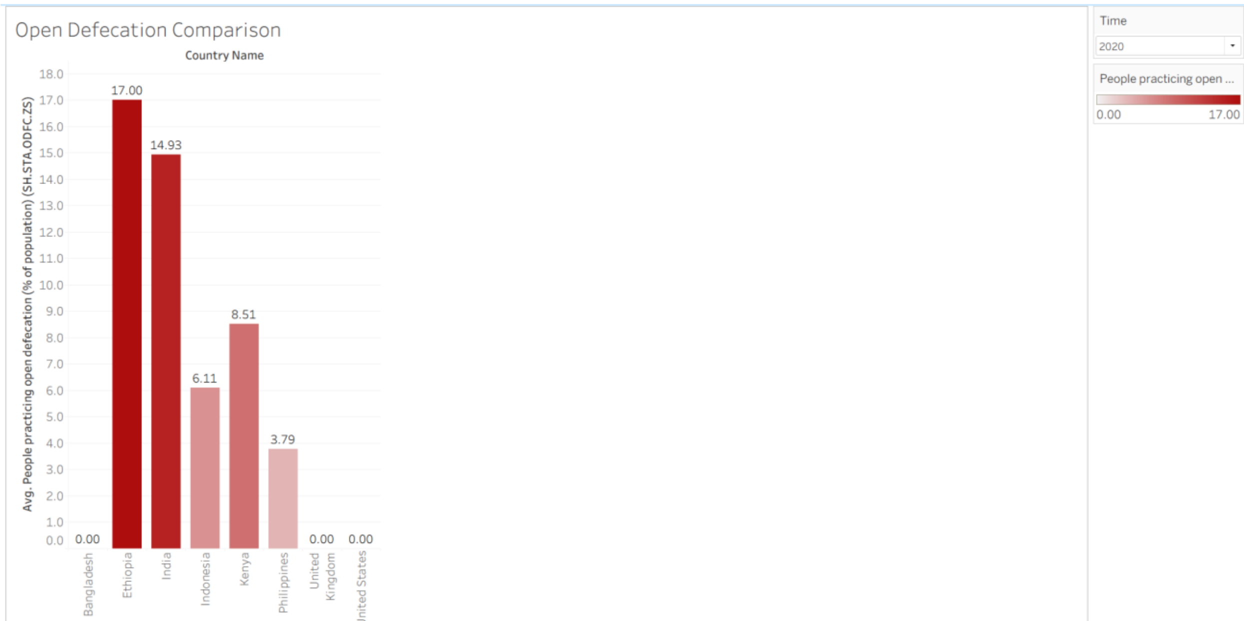
3.5 Water Access vs Under Five Mortality

This scatter plot conveys the message that the higher water access leads to lower child mortality. Developed nations like UK and USA cluster together with high water access and low mortality rate i.e. less

than 10 per 1000 whereas developing nations like Ethiopia and Kenya indicates lower water access leading to **higher child mortality**.

This is one of the strongest visuals that directly conveys the message of water related interventions for strategic decision-making. For Unilever, this **correlation between mortality rate and water access** is quite significant as it directly impacts the long-term sustainability of its consumer markets.

3.6 Open Defecation Comparison



Through this visual we can verify the importance of sanitation infrastructure investment. Countries like UK and USA show rates near zero whereas Ethiopia and India ranks highest due to the lack of sanitation infrastructure.

This visual is linked with many aspects as open defecation leads to the contamination of water source, which can further lead to childhood disease and mortality. Unilever should focus on **campaigns** like **Lifebuoy "Help a Child Reach 5"** initiative which focuses on educating people about washing hands and sanitation.

4.Recommendations

Through this analysis we can see multiple opportunities for Unilever to contribute more towards the SDG 3 AND SDG 6. Firstly, Unilever should focus on countries like Ethiopia, Bangladesh, Kenya and India where water security and sanitation infrastructure is needed. These regions are suffering due to low water access, High open defecation, and higher under-five mortality rate. A stable partnership with governments, NGOs, and community will help Unilever to accelerate progress.

Secondly, Unilever should focus on making products that supports the areas with low water access. Products like low-rinse detergent, waterless personal-care products, and affordable solutions for filtering water to help the communities will strengthen Unilever's market position and create social impact globally.

Thirdly, Unilever should focus on campaigns that can help people understand how sanitation can improve the lifestyle and also help in preventing diseases that can be prevented. Community-led sanitation programs supporting UN's SDG 3 and SDG 6 goals will also help Unilever gain customer base in countries where hygiene promotion is needed.

Fourth, Unilever should include BI dashboards in routine sustainability evaluation, which will significantly improve decision making. They should integrate quarterly evaluation dashboards, which can lead to stronger decision making. The dashboard which we have used as reference can benefit Unilever for analyzing the upcoming trends, so that they can prepare for optimized resource allocation. BI-driven insights are useful for long-term plans as it helps us identify trends and patterns that can hinder the future operations.

5.Critical Evaluation Insights Using BI

Working on this assignment helped me understand the importance of Business Intelligence. It is not about building dashboards; it is about the story you are able to convey with your data. By applying Business Intelligence, I understood that the first step is defining what decision the dashboard must support. In my analysis, I focused on how different countries, have varying levels of access to clean water and child health which aligns with the SDG 3 and SDG 6 goals.

With the help of Business Intelligence, I gained insights of the data reliability of world bank dataset. By analyzing the completeness, accuracy and details, I understood where the gaps existed which helped me filter accurate insights. The dashboards reflect descriptive analysis, but causation cannot be inferred.

Overall, the evaluation shows that BI tools help gain valuable insights and contextual understanding about the SDG goals and how, Unilever can seize the opportunity to become unique in the global market.

6.Conclusion

This report brought together the global SDG data, BI Analysis and Unilever's goal towards sustainability shows how data-driven decision making aligns with the societal needs. Through the dashboards, the insights gained can be used to analyze the emerging markets for organizations which can lead to better investments and sustainable environment.

It visualizes the long-term opportunities by converting the complex indicators to meaningful insights. For Unilever, these insights support activities such as water-efficient supply chains, water-independent products and community hygiene campaigns. Improving the environment for children under-five not only support future market but also helps secure the future workforce and customer base.

Overall, BI is not only about excel and dashboards, it is about creating a strategic mechanism, that helps organizations understand social sustainability goals, strategic decision making and helps monitor long-term impact. When used efficiently, Business Intelligence can help organizations take decisions which benefit both business performance and global development goals.

7.References

World Bank DataBank
World Bank (2024) *Sustainable Development Goals (SDGs) DataBank*. Available at: https://databank.worldbank.org/source/sustainable-development-goals-(sdgs) (Accessed: [21/11/25]).

United Nations SDG Official Website
United Nations (2024) *Sustainable Development Goals*. Available at: https://www.un.org/sustainabledevelopment/sustainable-development-goals/ (Accessed: [21/11/25]).

Unilever Sustainability Reports
Unilever (2024) *Sustainability Strategy and Progress Reports*. Available at: https://www.unilever.co.uk/sustainability/ (Accessed: [21/11/25]).

Lifebuoy
Lifebuoy. (n.d.) *Our Mission*. [online] Available at: https://www.lifebuoy.com/global/our-mission.html [Accessed 25 Nov. 2025].

University of Glasgow Moodle (Course Material)
University of Glasgow (2025) *MGT5491 Business Intelligence – Course Folder*, Moodle. Available at: https://moodle.gla.ac.uk/mod/folder/view.php?id=5685444 (Accessed: [23/11/25]).